GWS ENGINEERING & INDUSTRIAL SUPPLIES LTD.
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CE 0343/16

0343/CPR/SHA/BJG6041035/02

QUALITY CERTIFICATE EN 10204 3.1

REPORT NO.:YF26GW515-56005

DATE（日期):JUN.2026

PURCHASER(客户):G.W.S ENGINEERING & INDUSTRIAL SUP	ISO NO .(ISO号码):ISO 9001 – 0070445	CONTRACT NO.(合同号):PO000002580-AUG	SIZE(规格):M16X60
MANUFACTURING DATE(生产日期):MAY.2026	QUANTITIES(数量):6400 PCS	MATERIAL(材料):10B33	FINISH(色别):HDG
COMMODITY(品名):EN15048-8.8/10Z-Non- Preloaded structural bolting assemblies		LOT NO.(批号):3043156005	

DIMENSIONAL INSPECTION(外形检测)

HEX BOLT-ISO4017-M16X60-8.8

SAMPLING DATE :MAY.2026 SAMPLES : 8 PCS

INSPECTION ITEM (检测项目)	STANDARD （标准值)	ACTUAL RESULT （实测值）	RESULT
APPEARANCE外观	OK	OK	OK
HEAD MARK 标记	XYLX-8.8SB	OK	OK
WIDTH ACROSS FLATS 对边	23.67-24	23.71-23.8	OK
HEAD HEIGHT 头部厚度	9.82-10.18	9.87-9.92	OK
BODY DIAMETER 杆径	F/T	F/T	OK
LENGTH 长度	59.4-60.6	59.67-60.37	OK
THREAD LENGTH 螺纹长度	F/T	F/T	OK
GO-GAGE before coating	6g/T	GO	OK
NO-GO-GAGE before coating	6g/Z	NO GO	OK
STRAIGHTNESS直线度	0.2MAX	OK	OK
ZINC THICKNESS锌层厚度um	50MIN	54.1-58	OK

HEX NUT-ISO4032-M16-10Z

SAMPLING DATE :MAY.2026 SAMPLES : 8 PCS

INSPECTION ITEM (检测项目)	STANDARD （标准值)	ACTUAL RESULT (实测值)	RESULT
APPEARANCE外观	OK	OK	OK
HEAD MARK 标记	XYLX-10ZSB	OK	OK
WIDTH ACROSS FLATS 对边	23.67-24	23.86-23.93	OK
HEAD HEIGHT 头部厚度	14.1-14.8	14.12-14.4	OK
ZINC THICKNESS锌层厚度um	50MIN	55.9-59.7	OK

WASHER-ISO7091-100HV

SAMPLING DATE :MAY.2026 SAMPLES: 8 PCS

Inspection item	Standard	Actual result	Result
Marking	XYLX-SB	OK	OK
Hole diameter(d1)	17.5-17.93	17.62-17.63	OK
Outer diameter (d2)	28.7-30	29.03-29.43	OK
Thickness(h)	2.4-3.6	2.75-2.97	OK

MECHANICAL PROPERTIES(机械性能检测)

HEX BOLT-ISO 898-1

SAMPLING DATE :MAY.2026 SAMPLES : 8 PCS

TEST ITEM 检测项目	STANDARD （标准值)	ACTUAL RESULT (实测值)	RESULT
HARDNESS硬度(HRC)	23-34	28-30.5	OK
		27-28.5	OK
TENSILE STRENGTH 抗拉强度(MPa)	830MIN	960-967	OK
YIELD STRENGTH屈服强度(MPa)	660MIN	841-859	OK
ELONGATION(延伸率）	12MIN	15	OK
REDUCTION AREA (缩面率)	52MIN	56.2	OK
PROOF LOAD(保载）(KN)	94.5	94.5	OK
Impact strenght冲击功KV[J]	27MIN	86/82.6/86.1	OK

HEX NUT-ISO 898-2

SAMPLING DATE :MAY.2026 SAMPLES : 8 PCS

TEST ITEM 检测项目	STANDARD （标准值)	ACTUAL RESULT (实测值)	RESULT
HARDNESS硬度(HRC)	36.9MAX	29-30	OK
PROOF LOAD(保载）(KN)	164.9	164.9	OK

WASHER-ISO7091

SAMPLING DATE :MAY.2026 SAMPLES: 8 PCS

TEST ITEM	STANDARD	ACTUAL RESULT	RESULT
Hareness(HV)	100HV	134-140	OK

SUITABILITY TEST组合实验-EN 15048-1/2

SAMPLING DATE :MAY.2026 SAMPLES : 8 PCS

TENSILE RESISTANCE OF BOLT/NUT ASSEMBLIES(KN)	130MIN	139.07-143.97	OK

CHEMICAL COMPOSITION (化学成分检测)

SAMPLING DATE :MAY.2026

Item	Material	Heat No.	C	Mn	P	S	SI		
BOLT	10B33	B1-2604092	0.34	0.76	0.016	0.009	0.2		
NUTS	35K	G533011927	0.34	0.74	0.014	0.002	0.14		
Washer	A3	EA62200436	0.05	0.46	0.012	0.001	0.21		

(SIGNATURE)

We certify that the information detailed above has been reproduced from the original test certificate furnished by the manufacturer.

Niall Crowley, Quality Manager

江苏甬怡紧固件有限公司 质量部 INSPECTION DEPARTMENT